8.5

$$\mu\left(\frac{d^2u}{dy^2}\right) = \frac{dP}{dx} = \text{Const}$$

a). $\mu\left(\frac{d^2u}{dy^2}\right) = \frac{dP}{dx} = \text{Const}$ 이므로

$$\frac{d^2u}{dy^2} = \frac{1}{\mu}\frac{dP}{dx}$$

$$\frac{du}{dy} = \frac{1}{\mu}\frac{dP}{dx}y + C_1$$

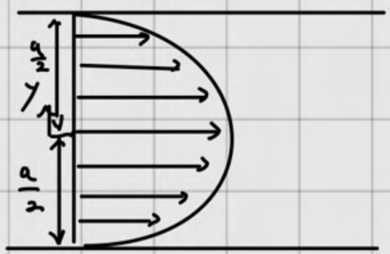
$$u = \frac{1}{2\mu}\frac{dP}{dx}y^2 + C_1y + C_2$$ 이때 $y=-\frac{a}{2}, u=0$ (No slip condition)

$y=\frac{a}{2}, u=0$

$$\therefore u = \frac{1}{2\mu}\frac{dP}{dx}y^2 - \frac{1}{2\mu}\frac{dP}{dx}\left(\frac{a}{2}\right)^2$$

$$= \frac{3}{2}\times\left(-\frac{a^2}{12\mu}\left(\frac{dP}{dx}\right)\right)\left(1-\frac{4}{a^2}y^2\right)$$

$$= \frac{3}{2}u_m\left(1-\frac{y^2}{(a/2)^2}\right)$$




b) $D_h = \frac{4A}{P}$ 로 계산된다.. $P$ : perimeter

$$= \frac{4(ab)}{2a+2b}$$

$b\to\infty$ 이라면

$$= \frac{4a}{\frac{2a}{b}+2} = 2a$$

$$\therefore f = \frac{-\frac{dP}{dx}D_h}{\rho u_m^2/2}$$

c). a)에서 얻은 속도분포를 활용하면

$$f = -\frac{dP}{dx}\frac{2D_h}{\rho u_m^2}$$

$$= -\frac{dP}{dx}\frac{2D_h}{-\frac{a^2}{12\mu}\left(\frac{dP}{dx}\right)\rho u_m}$$

$$= \frac{24\nu D_h^2}{a^2\ u_m D_h}$$

$$= \frac{24D_h^2}{a^2 Re_{D_h}} = \frac{96}{Re_{D_h}}$$ $\therefore C=96$

d) $u_m = -\frac{a^2}{12\mu}\left(\frac{dP}{dx}\right)$

$$= \frac{a^2}{12\mu}\frac{\Delta P}{\Delta x}$$

$$= \frac{(0.005)^2}{12\times 6\times10^{-7}} \times \frac{3.75}{0.2}$$

$$= 2.12\ m/s$$

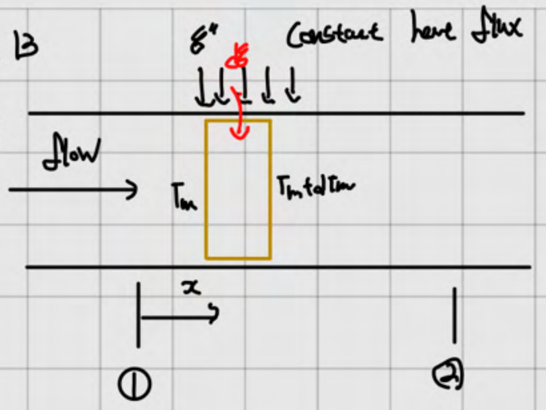
$$\therefore Re_{D_h} = \frac{u_m D_h}{\nu} = \frac{2.12\times 2\times 0.005}{15.89\times10^{-6}} = 1332$$

$\therefore$ laminar flow $\quad\therefore \frac{L}{d} = 0.05\,Re_{D_h}$

$$L = 0.05\times1332\times2\times0.005 = 0.67m$$

8.13

a)




CV에서

$\frac{dE}{dt} = dq + \dot m h_{in} - \dot m h_{out}$

$0 = dq + \dot m \left( c_p T_m - c_p (T_m + dT_m) \right)$

$\therefore dq = \dot m c_p dT_m$

이때 $dq = q'' P dx$

$\therefore \ q'' P dx = \dot m c_p dT_m$

$q'' = const$ 이므로 + constant properties

$T_m(x) = \frac{q'' P}{\dot m c_p} x + C_1 \quad (x = 0, \ T_m = T_{m,i})$

$\therefore T_m(x) = \frac{q'' P}{\dot m c_p} x + T_{m,i}$

∴ $T_m$은 직선 분포를 보인다.

CV로 대류 열전달 이므로

$q'' = h(T_s - T_m)$

Case A : thermally fully developed.

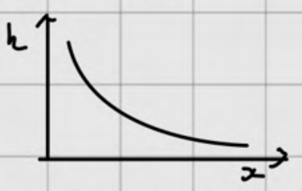
h가 바뀌지 않는다.

$\therefore \ \frac{q''}{h} = const$

$T_m + \frac{q''}{h} = T_s$

∴ $T_s$는 $T_m$과 $\frac{q''}{h}$ 만큼 차이나는 직선분포

Case B : entrance



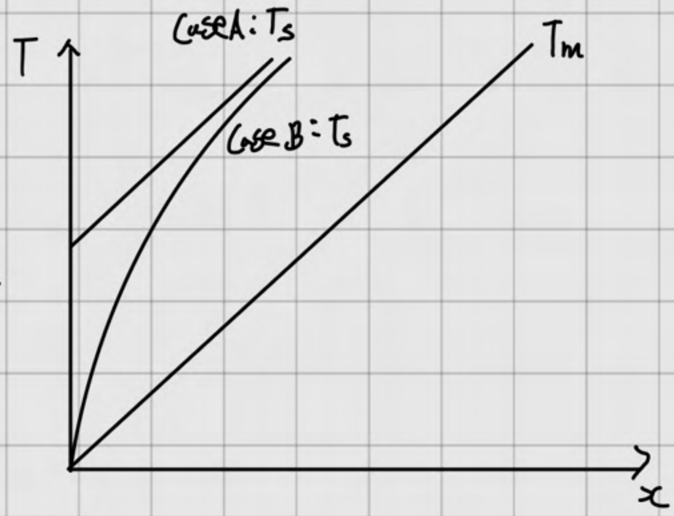

의 형태를 가지므로 $T_m + \frac{q''}{h} = T_s$ 에서

$T_s$는 $T_m$과 1영의 차이를 가짐

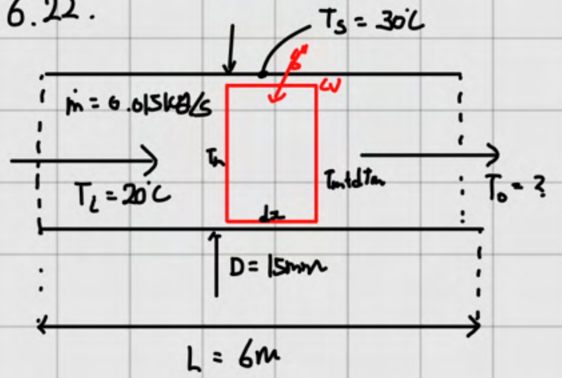
b) Case B의 경우 $T_s$, $T_m$의 차이가 1영로 나타나며 그 수렴값이 Case A의 차이가 되므로 Case A의 열전달이 더 크다.

따라서 $T_{m,2,A} > T_{m,2,B}$

∴



8.22.




Constant Surface Temperature, Const Properties

a) CV에 대해

$$\frac{dE}{dt} = Q - W$$

$$0 = q'' \pi D dx + \dot{m} c_p T_m - \dot{m} c_p (T_m + dT_m)$$

$$\frac{dT_m}{dx} = \frac{q'' \pi D}{\dot{m} c_p}$$

$$\frac{dT_m}{dx} = \frac{\pi D}{\dot{m} c_p} h (T_s - T_m) \quad \text{let } \theta = T_s - T_m$$

$$\frac{d\theta}{dx} = -\frac{\pi D}{\dot{m} c_p} \theta$$

$$\int_{\theta_i}^{\theta_o} \frac{d\theta}{\theta} = \int_0^6 -\frac{\pi D h}{\dot{m} c_p} dx$$

$$\ln \frac{\theta_o}{\theta_i} = -h \frac{6\pi D}{\dot{m} c_p} \quad \cdots (1)$$

$$\dot{m} = \rho A V$$

$$V = \frac{\dot{m}}{\rho A}$$

$$\therefore Re_D = \frac{VD}{\nu} = \frac{\dot{m} D}{A\mu} = \frac{4\dot{m}}{\pi D \mu} = 1489.$$

∴ laminar

$$\frac{z}{D} = 0.05 \, Re_D Pr = 432$$

$$\therefore z = 432D = 6.5m$$

∴ 전영역 entrance region

이때의 $$\overline{Nu_D} = \frac{\dfrac{3.36}{\tanh(2.264 Gz_D^{-1/3} + 1.7 Gz_D^{-2/3})} + 0.0499 Gz_D \tanh(Gz_D^{-1})}{\tanh(2.432 Pr^{1/6} Gz_D^{-1/6})}$$

$$\therefore \bar{h} = 203 W/m^2 \cdot K$$

(1)에 대입하면

$$\ln \frac{\theta_o}{\theta_i} = -h \frac{6\pi D}{\dot{m} c_p}$$

$$\theta_o = \theta_i \exp\left(-h \frac{6\pi D}{\dot{m} c_p}\right)$$

$$= 10 \exp\left(-203 \times \frac{6 \times \pi \times 0.015}{0.015 \times 4200}\right)$$

$$\theta_o = 4.02$$

$$\therefore T_{m,o} = 26$$

$$\therefore q = \dot{m} c_p (T_o - T_i)$$

$$= 378 W$$

b) for nanofluids

$k = 0.705\,W/m\cdot K$ . $\rho = 1146\,kg/m^3$ $c_p = 3.587\,kJ/kg\cdot K$ , $\mu = 962 \times 10^{-6}\,N\cdot s/m^2$ , $\alpha = 171 \times 10^{-9}\,m^2/s$

$\dot{m} = \rho A V = 0.015\,kg/s$

$\therefore V = \dfrac{0.015}{\rho A} = 0.074\,m/s$

$\therefore Re_D = \dfrac{VD}{\nu} = 1322$ (laminar)

$Pr = \dfrac{\nu}{\alpha} = 4.91$

$\dfrac{x}{D} = 0.05\,Re_D\,Pr$

$x = 4.87\,m$

$Gz_D = \left(\dfrac{D}{x}\right) Re_D\,Pr = \dfrac{15 \times 10^{-3}}{4.87} \times 1322 \times 4.91 = 19.99$

Combined entry에 대해 $\overline{Nu_D} = \dfrac{\dfrac{3.36}{\tanh(2.264\,Gz_D^{-1/3} + 1.7\,Gz_D^{-2/3})} + 0.0499\,Gz_D \tanh(Gz_D^{-1})}{\tanh(2.432\,Pr^{1/6}\,Gz_D^{-1/6})}$

$\therefore \bar{h} = 220\,W/m^2\cdot K$

$\therefore \ln\dfrac{\theta_o}{\theta_i} = -h\dfrac{6\pi D}{\dot{m} c_p}$

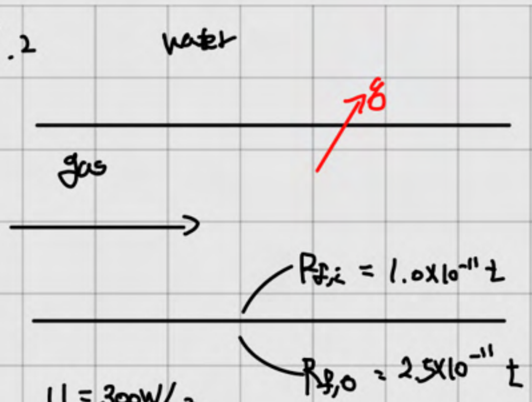
$= 10\exp(-220 \times 6 \times \pi \times 15 \times 10^{-3} \div 0.015 \div 3587)$

$\theta_o = 3.147$

$\therefore T_{m,o} = 26.8\,^\circ C$

$\therefore q = \dot{m} c_p (T_{m,o} - T_{m,i}) = 365\,W$

11.2



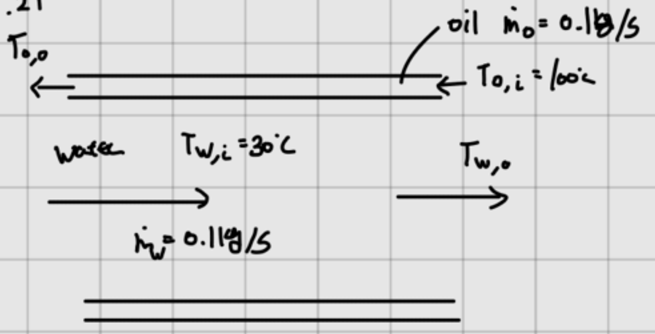

$U = 300\,W/m^2$

$$\frac{1}{U_{clean}} = \frac{1}{U} + R_{f,i} + R_{f,o}$$

$$\frac{1}{0.75\times300} = \frac{1}{300} + a_i t + a_f t$$

$$\therefore t = \frac{\frac{1}{300\times0.75} - \frac{1}{300}}{a_i + a_f} = 31746031\ S$$

11.21




$d_o = 45mm$

$d_i = 25m$

$U = 60W/m^2$

a) If $T_{o,o} = 60°C$ $q = ?$, $T_{w,o} = ?$

$q = \dot{m}C_{p,o}T_{o,o} - \dot{m}C_{p,o}T_{o,i}$

$= \dot{m}C_{p,o}(T_{o,o} - T_{o,i})$

$= (0.1)(1900)(-40)$

$= -7600 J$ from oil

then for water

$q = \dot{m}C_{p,w}(T_{w,o} - T_{w,i})$

$7600 = (0.1)(4200)(T_{w,o} - 30)$

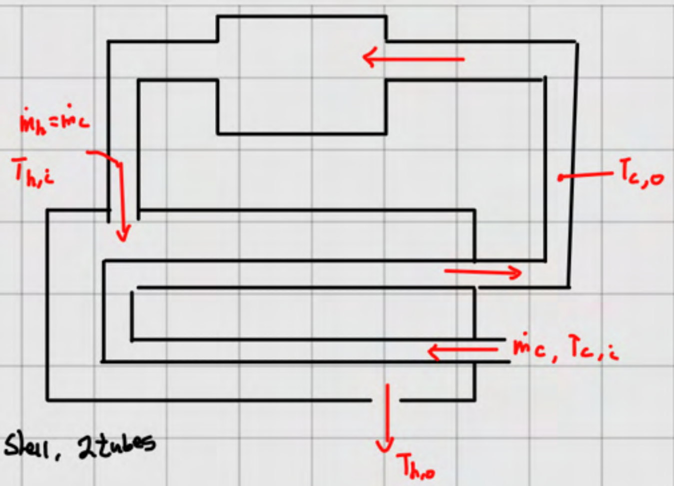
$\therefore T_w = 48.1°C$

b) $q = \pi D_i \times L \times q'' = \pi D_i L U \Delta T_{lm} = 7600$

$$\Delta T_{lm} = \frac{\Delta T_1 - \Delta T_2}{\ln \frac{\Delta T_1}{\Delta T_2}} = \frac{51.9 - 30}{\ln \frac{51.9}{30}} = 39.95$$

$$\therefore L = \frac{q}{\pi D_i U \Delta T_{lm}} = \frac{7600}{\pi (25 \times 10^{-3})(60)(39.95)} = 40.4m$$

11.24.




1 shell, 2tubes

$T_{c,i} = 5C$

$T_{h,i} = 70°C$

$\rho = 1030 \, kg/m^3$

$c_p = 3860 \, J/kg \cdot K$

a) $\dot{m} = 5 \, kg/s$, $U = 2000 \, W/m^2 \cdot K$, $\varepsilon = 0.5$, $A = ?$, $T_o = ?$, $q = ?$

$C_c = \dot{m} c_p = 19.3 \, kJ/K$ $\quad \therefore C_r = 1$

$C_h = \dot{m} c_p =$ "

$$\therefore E = \frac{2/\varepsilon - (1+C_r)}{(1+C_r^2)^{1/2}} = 1.414$$

$$\therefore NTU = -(1+C_r^2)^{-1/2} \ln\left(\frac{E-1}{E+1}\right)$$

$$\frac{UA}{C} = 1.2468$$

$$\therefore A = \frac{14300}{2000} \times 1.2468 = 12.03 \, m^2$$

$$q = \varepsilon C (T_{h,i} - T_{c,i})$$
$$= 627250 W$$

$$\therefore q = C(T_{c,o} - T_{c,i})$$

$$T_{c,o} = 37.5 °C$$

$$q = C(T_{h,i} - T_{h,o})$$

$$T_{h,o} = 37.5 °C$$

b) $\eta = 0.9$, $C_{ng} = \$ 0.02/MJ$ $\quad$ Annual Savings = ?

by installization we get heat recovery rate $q = 627250 J$

$$\therefore \Delta E = Q - \cancel{W} = q \times 3600 \times 24 \times 365 = 1.978 \times 10^{13} J$$
$$= 1.978 \times 10^7 MJ$$

$$\therefore \text{Annual savings} = \frac{\Delta E \times 0.02/MJ}{0.9} = 439576 \$$$